

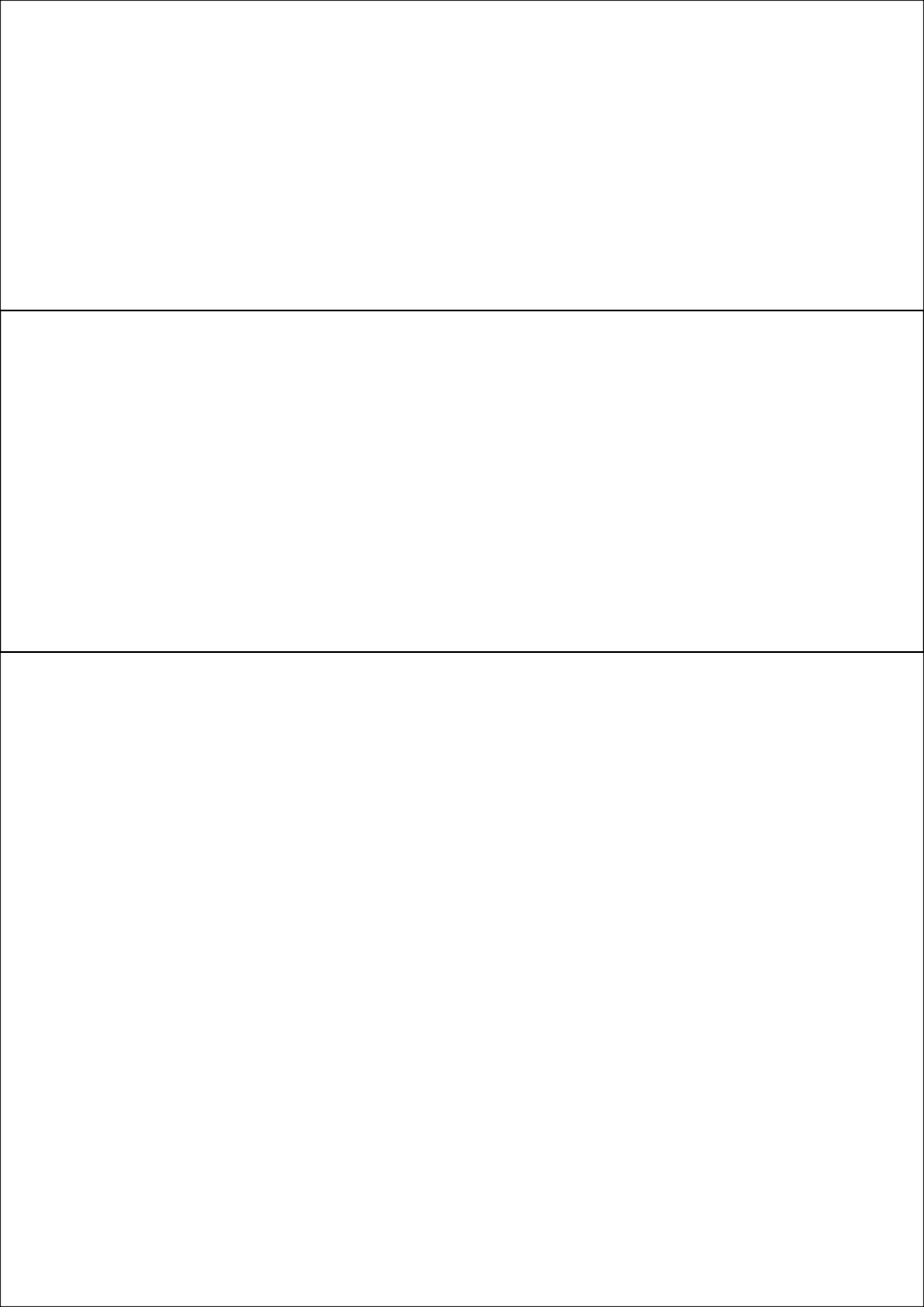


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1. Overview

GlobalPath is an online platform designed to support international students and immigrants throughout their journey abroad. It provides AI-powered visa guidance, a verified housing marketplace, mentorship connections, job listings with visa-sponsor filtering, and a comprehensive life-support toolkit.

The platform was developed as a Final Year Project (FYP) by Group 8 during the 2024/2025 academic year at KNUST. It addresses the critical gap between admission and settlement that traditional platforms like Common App or LinkedIn do not cover — offering end-to-end support from visa application through to post-arrival integration.

Key problems it solves: fake visa agents defrauding students, unverified housing scams, lack of structured mentorship for diaspora-bound students, and difficulty finding visa-sponsoring employers.

2. Team

Member	Role	Owns
Eric Asante (3376122)	Backend & Database	backend/, ai/, db/, deployment
Baddoo Jeremiah Nii Adotei (3381622)	Frontend & UI/UX	frontend/, UI/UX, design system

3. Architecture

globalpath/
% % % frontend/ Next.js 15 + Tailwind v4 (React UI, SSR)
% % % backend/ Node.js + Express (REST API + WebSocket)
% % % ai/ Python + FastAPI (Visa assistant, doc checker)
% % % db/ PostgreSQL schema + migrations
% % % docker-compose.yml Postgres + Redis local

Request flow:

[Browser]
!"

[Next.js :3000] % % rewrites /api/*% % !" [Express :4000]
!" uses
[Postgres :5432]
[Redis :6379]
!" proxies AI
[FastAPI :8000]
!" uses
[Anthropic Claude]
[Google Translate]

4. Tech Stack

Layer	Tech
Frontend	React.js / Next.js 15 (App Router)
Backend API	Node.js / Express (REST + WebSocket)
Database	PostgreSQL
Cache	Redis

AI Engine	Python / FastAPI
Auth	JWT / OAuth 2.0
Translation	Google Translate API
Hosting	Vercel (frontend) + AWS/Heroku (backend)
Storage	AWS S3 / Cloudinary
Payments	Stripe / Pystack
Email/SMS	SendGrid / Twilio

5. Services and Ports

Each service runs on its own port during development:

- Frontend — <http://localhost:3000>
- Backend API — <http://localhost:4000> (health: GET /health)
- AI Service — <http://localhost:8000> (health: GET /health)

6. Quick Start

Prerequisites: Node.js 20+, Python 3.11+, Docker (for Postgres + Redis).

Step 1: Start infrastructure

```
docker compose up -d
```

This boots Postgres 16 + Redis 7 containers. Schema auto-loads from db/schema.sql.

Step 2: Backend

```
cd backend
Copy-Item .env.example .env # Windows
npm install
npm run dev
```

Step 3: AI Service

```
cd ai
python -m venv .venv && .venv\Scripts\Activate.ps1 # Windows
pip install -r requirements.txt
Copy-Item .env.example .env
uvicorn main:app --reload --port 8000
```

Step 4: Frontend

```
cd frontend
Copy-Item .env.example .env.local # Windows
npm install
npm run dev
```

Smoke Test

Open <http://localhost:3000> to see the landing page. Register a student account, land on dashboard, and try the AI Assistant in the sidebar.

7. Key Features

AI Visa Assistant

Conversational guidance powered by Anthropic Claude. Users describe their situation (origin, destination, visa type) and receive step-by-step instructions with personalized document checklists. The AI cites official government sources and includes disclaimers. Users can escalate to human mentors on complex cases.

Verified Housing Marketplace

Identity-checked landlords with multi-step verification: government ID, proof of ownership, video walk-through, and positive reviews from verified tenants. Roommate matching via lifestyle preferences. Deposits held in Stripe Connect escrow.

Mentorship Network

Connect with verified diaspora mentors who have lived in the target destination. Filter by origin-destination pair, field of study, or profession. Free 1:1 video sessions for verified students. Built-in calendar and meeting link system.

Jobs & Internships

Filter listings by visa-sponsorship history. AI-driven resume builder with real-time preview. Sponsorship-history tracker showing actual sponsored counts from past 12 months. Conditional applications accepted for students still obtaining permits.

Life Support Toolkit

Cost-of-living calculators across 50+ cities. Healthcare registration guides by country. Banking setup instructions (blocked accounts, student accounts). Emergency SOS system with one-tap alerts to contacts, embassy, and student union.

Scam Alert System

Community-reported and admin-verified fraud warnings. Categories: visa scams, housing fraud, phishing, job scams. Red 'Report' button on every listing and DM. Admin team action within 4 hours on verified scams.

Additional Features

- AI Application Coach — scores essays against rubric, suggests improvements
- Document Validity Checker — analyzes uploaded docs for rejection triggers
- Scholarship Matcher — continuous scan of verified database against user profile
- Peer Review Essays — anonymous feedback from mentors with structured rubrics
- 50+ Language Support — auto-detect and translate via Google Translate API

8. AI Service Modules

File	Purpose
ai/main.py	FastAPI app with /chat, /checklist, /doc-check, /translate
ai/visa_assistant.py	Claude-powered conversational visa guidance
ai/checklist_generator.py	Generates personalized visa document checklists
ai/doc_checker.py	Analyzes documents for common rejection triggers
ai/translator.py	Multi-language translation service
The AI service uses Anthropic Claude with prompt caching for cost control. All endpoints are rate-limited. The assistant operates in stub mode when no ANTHROPIC_API_KEY is set, providing generic	

visa flow guidance.

9. Database

PostgreSQL 16 with the following schema structure:

Schema (db/schema.sql):

- users + auth (roles: student, mentor, employer, admin)
- mentor_profiles, employer_profiles
- opportunities (scholarship, work_study, exchange, internship, job)
- housing_listings + roommate_preferences
- forum_categories + forum_posts + forum_replies
- conversations + messages
- ai_conversations + ai_messages + visa_checklists
- reports + scam_alerts

Migration 002 (db/migration_002.sql):

- success_stories
- notifications
- saved_items
- mentor_bookings

Seed data (db/seed.sql) includes sample mentors (Ama Owusu, Kwame Adjei, Yaa Boateng, Priya Sharma, Tunde Adebayo), admin (Sarah Admin), 7 scholarship/job/exchange opportunities, 5 housing listings, and 4 scam alerts. All passwords use bcrypt hash of 'testpass123'.

10. Infrastructure & Deployment

Local Development

- Docker Compose runs Postgres 16 (container: globalpath_postgres) and Redis 7 (container: globalpath_redis)
- Database URL: postgresql://globalpath:globalpath_dev@localhost:5432/globalpath
- Redis URL: redis://localhost:6379

Production Targets

- Frontend: Vercel (vercel --prod from frontend/)
- Backend: Heroku, Railway, or AWS Elastic Beanstalk
- AI Service: AWS Fargate, Google Cloud Run, or Railway
- Database: Supabase, Neon, or AWS RDS
- Redis: Upstash or AWS ElastiCache
- Storage: Cloudinary or AWS S3
- Payments: Stripe / Paystack
- Email: SendGrid; SMS: Twilio

Third-Party Integrations

- Anthropic Claude — AI chat, document checking, essay scoring
- Google Translate API — 50+ language support
- Stripe / Paystack — payment processing
- SendGrid — transactional email delivery
- Twilio — SMS alerts and notifications
- Cloudinary — image upload and optimization

11. Branch Convention

- main — production-ready
- dev — integration branch
- feat/<scope> — per feature branches

Team workflow: Eric owns backend/, ai/, db/, and deployment. Jeremiah owns frontend/, UI/UX, and design system.